Brussels student housing - conditional research screening

Executive conclusion: no actual property purchase is recommended. The source spreadsheet is missing and none of the example sites has legal, pricing or condition verification. Synthetic A/B/C illustrate the decision-support method only.

Exactly150000 simulations completed:50000 per conditionally feasible example. An8m2 no-reconfiguration exampleD is excluded before simulation. Financial portfolio score selects No investment under the assumed risk and funding constraints.

The analysis emphasizes expected economic value, downside cash losses, peak capital at risk, time to letting and compliance conditions. It does not manufacture live opportunities or market-calibrated success probabilities.

Comparative investment ranking

A: {"trials": 50000, "mean": -954694.2608708977, "P5": -1213207.0819239472, "P50": -952932.8237957094, "P95": -700726.533393171, "P_negative": 1.0, "P_target": 0.0, "P_nominal_loss": 0.18518, "mean_profit": 544666.0304699469, "P95_funding": 1547698.7587493611, "CVaR5": 1277962.0091414503}

B: {"trials": 50000, "mean": -1146348.9755670996, "P5": -1502561.24588548, "P50": -1145200.4667999502, "P95": -796458.6351194727, "P_negative": 1.0, "P_target": 0.0, "P_nominal_loss": 0.06422, "mean_profit": 1242260.27869828, "P95_funding": 1877811.3054972903, "CVaR5": 1589708.1966336092}

C: {"trials": 50000, "mean": -1295679.2717979902, "P5": -1744974.5380283515, "P50": -1292869.6168830143, "P95": -859359.7673377057, "P_negative": 1.0, "P_target": 0.0, "P_nominal_loss": 0.03088, "mean_profit": 1895779.6881812888, "P95_funding": 2229479.275988769, "CVaR5": 1854260.027160081}

Metrics are equity after simplified corporate/gain taxes. P(target) is nonnegative equity NPV at10% hurdle. Positive expected NPV alone does not imply acceptable downside or legal approval.

Financial projections and timing

Acquisition, works, fees, assumed duty and VAT, furniture and liquidity are funded at time0. Annual rent, variable and fixed costs, taxes, replacement capex, loan amortization and terminal property sale are separate.

Readiness is rounded to the next September letting cycle. A project ready after September may miss almost a full academic year. Opening month21 means only one quarter of year2 generates rent; horizon still ends at year20.

Standard corporate rate25% and general Brussels acquisition duty12.5% were verified from official sources. Depreciation lives, tax basis, company regime and interest deduction treatment are simplified assumptions; no loss carryforwards.

Property value is capitalized year21 NOI at simulated exit cap. Income growth and exit cap capture appreciation; an additional appreciation uplift is not double-counted. Debt is repaid once at exit.

Simulation, dependence and risk

Nine uncertain quantities vary simultaneously: acquisition cost, works, rent, occupancy, rate, readiness, exit cap, rent growth and operating cost. Seed29102026; all150000 input/output rows in Excel and CSV.

Triangular min/mode/max: acquisition0.95/1/1.1; works0.85/1/1.25; rent0.85/1/1.1; occupancy0.75/0.95/0.99; rate3/4.5/7%; readiness6/12/24 months; exit cap4.5/5.5/7.5%; growth0.5/2/3.5%; opex0.9/1/1.2. These are assumptions, not estimated distributions.

Shared cost/delay and credit/exit factors create assumed cross-property dependence. Portfolio outcomes use paired same-trial values rather than adding unrelated percentiles. All valid negative outcomes retained.

Main dynamic Excel cash-flow formulas are checked; simulation and sensitivity outputs are static and must be regenerated from the supplied source script after changing input definitions.

Optimization and managerial decision

['No investment', 0, 0, 0.0, True, 0.0, 0.0, 'No acquisition']

['A', -954694.2608708977, 1277962.0091414503, 1547698.7587493611, True, -1274184.7631562604, 1.0, 'Unverified']

['B', -1146348.9755670996, 1589708.1966336092, 1877811.3054972903, True, -1543776.024725502, 1.0, 'Unverified']

['A+B', -2101043.2364379973, 2713062.256543218, 3342220.3464802178, False, -2779308.8005738016, 1.0, 'Unverified']

['C', -1295679.2717979902, 1854260.027160081, 2229479.275988769, True, -1759244.2785880105, 1.0, 'Unverified']

['A+C', -2250373.532668888, 2965415.101129441, 3698314.287913019, False, -2991727.3079512483, 1.0, 'Unverified']

['B+C', -2442028.24736509, 3234292.256875606, 4061700.013074379, False, -3250601.3115839916, 1.0, 'Unverified']

['A+B+C', -3396722.5082359877, 4346446.097225458, 5504228.523915927, False, -4483334.032542353, 1.0, 'Unverified']

Objective: mean equity NPV minus0.25 times the mean worst5% NPV loss. Constraint:95th percentile funding<=EUR3m. Both are investor assumptions. All8 subsets including no investment were evaluated.

Legal approval is mandatory. Under actual evidence available, no candidate is legally cleared; this overrides financial ranking. Dynamic reinvestment was not implemented.

Compare maximum supportable acquisition prices, readiness9 versus12 month outcomes and10/15/20-year horizons in the workbook before negotiating a real asset.

Sources and outstanding diligence

Brukot market asking rents: https://www.brukot.be/en/student-guide/rent-stats

2026 quality text: https://be.brussels/fr/logement/location/conformite-logement/normes-et-securite

Corporate tax: https://finances.belgium.be/fr/entreprises/impot_des_societes/declaration

Acquisition duty: https://be.brussels/en/taxes-finances/taxes/property-related-taxes/registration-duties

Ixelles planning: https://www.ixelles.be/site/81-Service-de-l-Urbanisme-et-de-l-Environnement

All checked29 October2026. Obtain actual spreadsheet, addresses, measured net areas, permitted use, fire/sanitary/PEB assessments, rent comparables, quotes, financing and tax advice.